

# MRI Digital Twin Handover

A generative model that invents a realistic month of activity for an MRI scanner site — which patients arrive, what gets scanned, in what order, and how long every step takes. This is where it stands, what it gets wrong, and what to do next.

Author Luke Schumacher    Date 31 August 2026    Repo Time-Series-Models @ 28ed58b

Status Working, with one blocking defect

|                       |                       |             |                       |                                    |
|-----------------------|-----------------------|-------------|-----------------------|------------------------------------|
| 3                     | 20                    | 395         | 51,321                | 344/346                            |
| TRANSFORMER<br>MODELS | SCANNERS<br>GENERATED | TESTS GREEN | TRAINING<br>SEQUENCES | MODEL INPUTS<br>THAT DO<br>NOTHING |

## 01 Where we are

READ THIS  
FIRST

Two of the three models are good enough to ship. The third — the one that predicts how long a scan takes — is a very expensive lookup table, and that is the whole story of the last two months.

### What works

The **patient-handover model** is the win column. Its event mix matches real scanners almost exactly, and the day-level volume is right: synthetic visits contain 7.9 scans on average against a real 8.6, with an identical median of 8. Scan lengths are the right order of magnitude, aborted scans appear at roughly the right rate (3.0% against a real 3.6%), and the whole pipeline runs end to end without hand-holding.

### What does not

Three things, in order of how much they matter:

- **The duration model ignores every input except the protocol name** — and the protocol name is the one input it is not allowed to use. It is beaten on held-out data by a spreadsheet.
- **Anatomy collapses.** Real scanners cover 56 body parts; we generate 7. Eight of the fifteen real body groups are never produced at all.
- **Scans run too long and visits have no pauses.** 4.5% of generated scans exceed ten minutes; in real data it is 0.06%. Downstream, the day is over-filled and the simulator throws away patients it had already scheduled.

## The single next move

Stop adding features to the duration model. Change what it is asked to predict: train it on the *residual* against the protocol average, or remove the protocol input entirely and score against it. The evidence that this will work is in §5 — fitted as a plain table on the same rows, the sequence parameters reach 9.7 s of error against the protocol benchmark's 13.2 s. The signal is in the data. The network is not using it.

### BEFORE YOU SHIP THE NEXT QLIK BUNDLE

The 40 CSVs in `~/Downloads/Results 31082026` are **two generator fixes behind the code**. Measured on those files: 100% of the gaps between scans inside a visit are exactly 0 s (commit `c9e8604` put that pause back), and the protocol column carries the old eight-name placeholder pool ( `t1_mprage` , `t2_flair` , ...) rather than sampled real protocol names. Their per-scanner row counts do not match the step-07 log from the latest run either — 3,793 rows for scanner 183242 here against 3,574 in the log.

They are perfectly usable for the anatomy and event-mix charts. Re-run step 07 on current `main` and re-download before using them to judge timing.

## 02

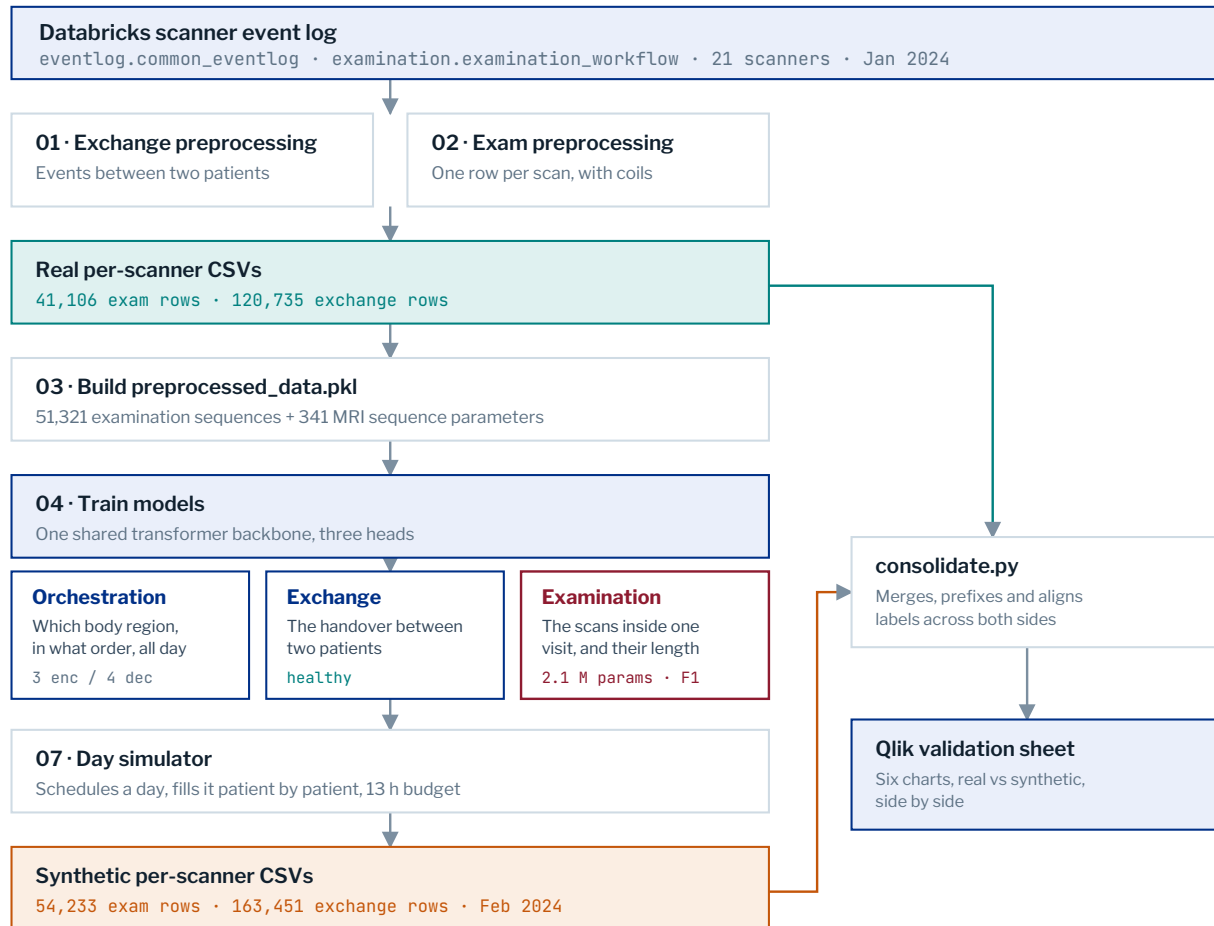
HOW IT IS  
BUILT

## Architecture

Real scanner event logs go in at the top; a month of invented-but-plausible scanner activity comes out at the bottom, in exactly the same CSV shape. That shape-match is deliberate: it is what lets the Qlik dashboard put real and synthetic side by side without a single column rename.

## The pipeline, top to bottom

Numbered boxes are Databricks notebooks. Teal = real data, orange = generated.



All three models are the same transformer: 128-dimensional, 4 attention heads, 3 encoder and 3 decoder layers, 512-wide feed-forward, with a log-space duration head. Small on purpose — the training set is tens of thousands of sequences, not millions.

## Two pipelines, and why

The repository carries the pipeline twice. `csv_pipeline/` is the original and owns the exchange and orchestration checkpoints plus the real-data extracts the dashboard compares against. `csv_pipeline_seqparams/` is a fork that adds 341 MRI sequence parameters — repetition time, slice count, field of view and so on, read out of the `MRI_SUT_1005` message — to the examination model's conditioning. It was forked rather than modified so a failed experiment could not take the working checkpoints down with it. That decision paid off; see §4.

## Why the protocol name is not allowed to be an input

This is the constraint that everything else follows from, set with Görtler on 31 July 2026. A protocol is a saved state — one sequence plus roughly fifty parameter

settings. Customers keep their own protocol trees, copy Siemens presets, rename them in any language, and edit them; copying destroys the link to the original preset. The result is around **3.8 million protocol names across 15,000 customers against only 3,800 Siemens presets**. A model keyed on the protocol name learns one site and transfers to none. So the name stays in the data as a *label* and becomes the benchmark to beat — never a feature.

03

The scoreboard

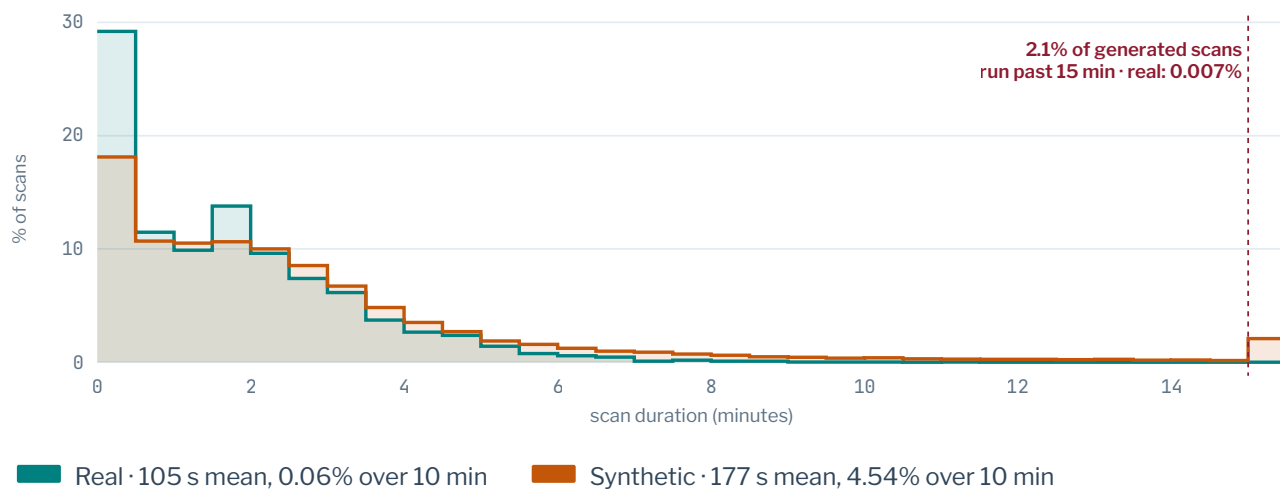
REAL VS  
SYNTHETIC

Every number below is measured from the CSVs on disk, not carried over from an earlier report. Real is 10 scanners over January 2024; synthetic is 20 scanners over February 2024, from the run currently in `~/Downloads/Results 31082026` .

| Measured comparison                                                                                                                                                                                                                                                                       |          |           |         |                                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------|---------|-----------------------------------|
| 41,106 real exam rows against 54,233 synthetic                                                                                                                                                                                                                                            |          |           |         |                                   |
| METRIC                                                                                                                                                                                                                                                                                    | REAL     | SYNTHETIC | VERDICT | NOTE                              |
| Scans per visit (mean)                                                                                                                                                                                                                                                                    | 8.6      | 7.9       | PASS    | Medians both 8                    |
| Aborted by user                                                                                                                                                                                                                                                                           | 3.61%    | 3.03%     | PASS    | Was 0% a year ago                 |
| Handover event mix, top 5                                                                                                                                                                                                                                                                 | —        | —         | PASS    | Same set, two ranks swapped       |
| Scan duration, median                                                                                                                                                                                                                                                                     | 88 s     | 120 s     | CLOSE   | +36%                              |
| Visit wall clock                                                                                                                                                                                                                                                                          | 22.7 min | 23.4 min  | CLOSE   | Right for the wrong reason        |
| Scan duration, mean                                                                                                                                                                                                                                                                       | 105 s    | 177 s     | FAIL    | +69%                              |
| Scans over 10 minutes                                                                                                                                                                                                                                                                     | 0.06%    | 4.54%     | FAIL    | 76× too many                      |
| Dead time inside a visit                                                                                                                                                                                                                                                                  | 33.5%    | 0.3%      | FAIL    | Fixed in code, not in these files |
| Handover block, mean                                                                                                                                                                                                                                                                      | 1,108 s  | 520 s     | FAIL    | Long tail lost; medians close     |
| Distinct body groups                                                                                                                                                                                                                                                                      | 15       | 7         | FAIL    | 8 never generated                 |
| Distinct body parts                                                                                                                                                                                                                                                                       | 56       | 7         | FAIL    |                                   |
| Distinct protocol names                                                                                                                                                                                                                                                                   | 3,000    | 8         | FAIL    | Placeholder pool in these files   |
| The wall clock matches by accident. A real visit is 15.1 min of scanning inside 22.7 min of elapsed time. Ours is 23.3 min of scanning inside 23.4 min. Scans are 54% too long and the pauses between them are zero, and the two errors happen to cancel. Do not read that row as a pass. |          |           |         |                                   |

## Scan duration, real against generated

Share of scans per half-minute, 41,106 real and 54,233 synthetic



The shapes agree between roughly one and four minutes. They part in both tails: real data has a hard wall at about seven minutes, and 40.7% of real scans finish inside a minute against 28.8% of ours.

## Four failures, named

Each one below has been reproduced and root-caused, not guessed at. The identifiers F1–F4 are referenced again in the ranked next steps in §6.

**F1**

### The duration model ignores every input except the protocol name

**BLOCKING**

#### WHAT WE SEE

Shuffle any one input and measure how far the prediction moves. Of 346 conditioning features, exactly two move it: the protocol name and the coarse scan type. The other 344 — every one of the 341 MRI sequence parameters, plus the body region and the scanner's own identity — move it by less than 0.05 s, which is indistinguishable from re-seeding the random number generator.

#### WHAT THAT COSTS

Held-out error is 19.1 s. A plain lookup table holding the average duration per protocol name scores 16.2 s on the same data. A 2.1-million-parameter transformer is losing to a spreadsheet — and it is losing while leaning on the one input it is not allowed to use in production.

#### WHY IT IS NOT A WIRING BUG

That was the first hypothesis and it was tested and rejected. The path that carries the conditioning into the duration head exists in the checkpoint and was trained — its weight norm is 1.16, against 1.07 for the scan-type path that demonstrably works. The channel is live and still dead.

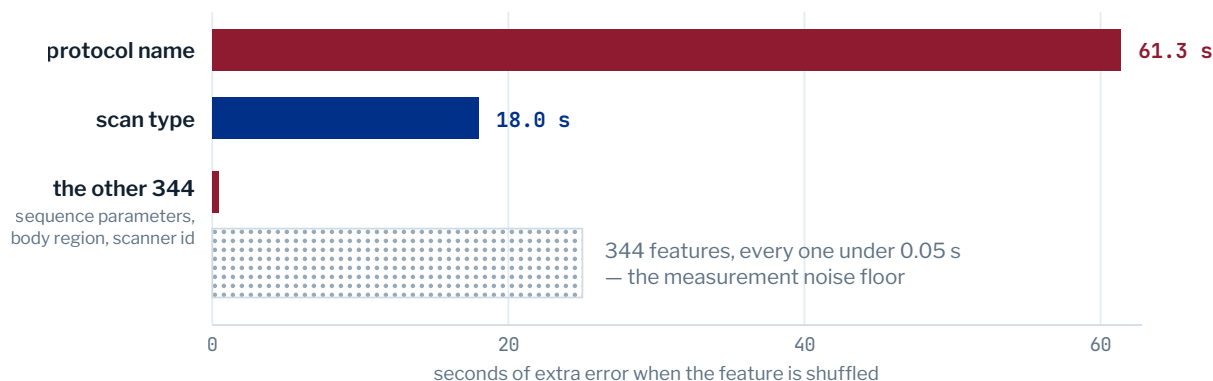
The likely mechanism is credit assignment, not capacity. A direct 1,620-row protocol lookup already explains 76% of held-out variance, so it is the cheapest way to reduce the loss, and the gradient reaching the 351-dimensional parameter path dies in the first few epochs. **Retraining the same objective will not fix this.**

#### WHY IT IS WORTH FIXING RATHER THAN ABANDONING

The signal is provably present. Fitted as an ordinary table on the same rows, the 89 admissible sequence parameters reach 9.7 s of error against the protocol benchmark's 13.2 s. That clears the  $\pm 15$  s target Görtler named, beats the benchmark, and uses nothing customer-specific. Roughly eight seconds of accuracy is sitting in the data and the network is not touching it.

## What the duration head actually listens to

Permutation sweep, 346 features × 5 repeats × 500 held-out scans



Each dot is one input feature. The two bars are the only channels the model responds to — and the long one is the channel that must be removed before this can ship to a second customer.

F2

## Anatomy collapses to seven categories

MAJOR

### WHAT WE SEE

Real scanners in this window cover 15 body groups and 56 body parts. We generate 7 and 7. `UNKNOWN` takes 19.5% of generated scans against 2.2% of real ones, and `HEAD` is over-produced at 40.5% against 25.5%. Knee (6.2% of real work), shoulder, breast, heart, hip, chest, other and whole-body are never generated at all.

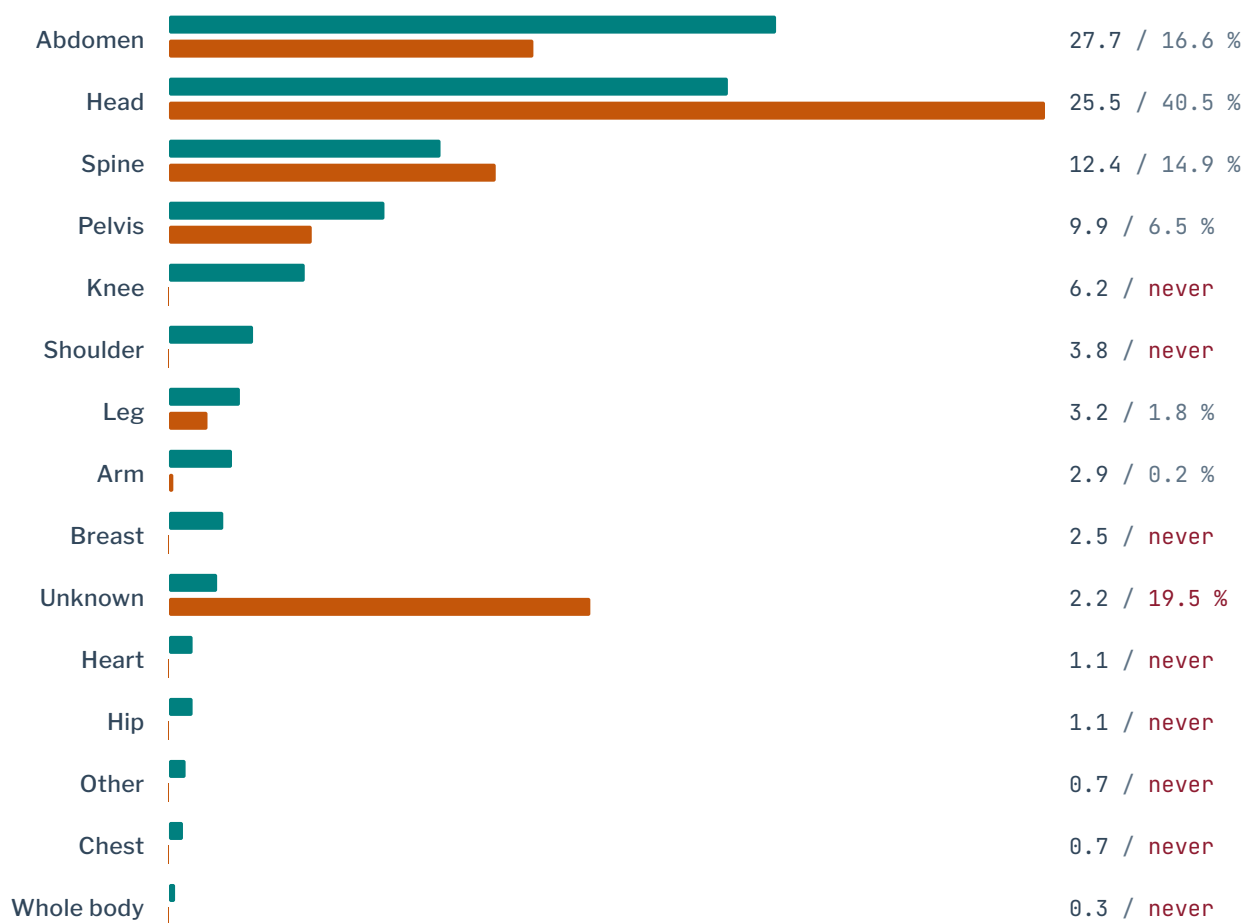
### ROOT CAUSE, IN TWO PARTS

**The vocabulary is too small.** The eleven-value body-region list in `config.py` has no knee and no heart, so step 03 folds brain, liver, knee, lumbar spine, prostate, shoulder and breast into `UNKNOWN` before training ever sees them.

**The source field is empty.** `BodyPartExamined` is null on nearly every exam row. Step 03 tries three guessed key names out of the workflow map and none of them is the real one. It now prints the keys that are actually present, so the next Spark run names the field and the fix is one line.

## Body-group share, real against generated

Sorted by real share · 15 groups real, 7 generated



Real Synthetic

**F3**

## Scans run long, and the day silently loses patients

**MAJOR**

### WHAT WE SEE

4.54% of generated scans exceed ten minutes; real data has 0.06%. The 99th percentile is 1,205 s against 392 s, and the longest generated scan runs 66 minutes against a real maximum of 19.

### THE DOWNSTREAM COST, WHICH IS THE REAL PROBLEM

The day simulator caps a scanner's day at 13 hours. Because generated scans are too long, that cap binds. In the most recent run it bound on **16 of the 20 scanners** and discarded **969 patients** — 13% of everything the orchestration model had scheduled — plus 938 individual scans. One scanner hit the cap on all 20 working days. Those patients do not appear in the output and nothing downstream knows they existed.

### THE PAUSE INSIDE A VISIT

A real visit is 15.1 minutes of scanning inside 22.7 minutes of elapsed time — a third of it is the operator repositioning, talking to the patient, changing coils. The generator advanced the clock by the scan duration alone, so the next scan started the instant the previous one ended. Commit `c9e8604` puts the pause back and the newest run reports 15–27% dead time against a real 30.4%. **The CSVs currently on disk predate that commit** — measured, 100% of their intra-visit gaps are exactly zero.

**F4**

## Patient handovers lose their long tail

**MINOR**

### WHAT WE SEE

The mean handover block is 520 s against a real 1,108 s, while the medians sit close together (381 s against 455 s). The model reproduces the typical handover and misses the rare very long one — a shift change, a lunch break, an equipment problem. The event mix itself is correct, which is why this is cosmetic rather than structural.

## One run settles F1

Do not add more features. Three attempts in a row added inputs — TR, slice count, then all 341 parameters — and the permutation result did not move once. The input is not the problem. The *target* is.

The duration head has a shortcut: predict the protocol's average and stop. Remove that shortcut from what it is asked to predict, and the parameters become the only route left.

### Two forms, pick one

1. **Residual training.** Subtract the protocol mean from the target, so the head learns only the part the operator changed — which is exactly the part the executed parameters describe. Add the mean back at generation time.
2. **Full ablation.** Build the model with no protocol tensors at all ( `num_protocols=None` ) and score it against the protocol oracle. Cleaner to interpret, and it is the configuration that would actually ship.

### The bar to clear

Held-out error below **16.2 s with the protocol name removed from the inputs**. Beating the previous checkpoint is not sufficient and never has been: a model can ignore a new input entirely and still look improved, which is precisely what happened when TR and slice count were added. The tabular fit says 9.7 s is reachable.

#### THE DIAGNOSTICS ARE ALREADY WRITTEN

`05_feature_analysis.py` prints the permutation ranking and  
`06_compare_models.py` prints the conditioning-path weight norm and branches on it — trained-but-dead means an objective problem, zero-initialised means a wiring problem. You do not need another exploratory cycle to know which branch you are in; one training run plus those two notebooks gives the answer.

### Why Görtler's argument makes this worth doing

If protocols ran exactly as saved, scan durations would vary by about a second. They vary by fifteen *because* the operator adjusts after loading — pushing slice count from 15 to 17 once the localiser shows the coverage, or dropping slices when a low-weight patient triggers a specific-absorption-rate limit. The protocol name is the label from before those adjustments. The parameters are what actually ran. Executed parameters should therefore beat the name, not merely match it — and our own data confirms the premise: within a single scanner-and-

protocol group, slice thickness varies on 80.5% of scans and repetition time on 56.7%. The message records what ran, not what was saved.

## 06

NEXT  
STEPS

# Ranked, with effort

## 1 Retrain the duration head on the residual target

The blocking defect. Everything else is polish until this moves.

**Effort** one GPU training run (~25 min) plus two analysis notebooks **Fixes** F1

## 2 Run step 03 with Spark and read the printed workflow keys

The step already enumerates the keys actually present in the workflow map. Reading that output names the field that should be feeding body part, and the fix is a one-line change. This is the cheapest large win on the board.

**Effort** ~4 min of Spark, then one line **Fixes** F2 at source

## 3 Widen the body-region vocabulary and retrain orchestration

Add knee, shoulder, breast, heart and hip to `BODY_REGIONS` and extend the step-03 mapping. Eight of fifteen real groups are currently impossible to generate — not unlikely, impossible.

**Effort** vocabulary edit plus one orchestration retrain **Fixes** F2

## 4 Regenerate and re-bundle for Qlik on current main

The shipped CSVs are behind the code by the intra-visit pause and the protocol sampler. Martina's dashboard is currently judging an older model than the one in the repository.

**Effort** one step-07 run plus `consolidate.py` **Fixes** F3 partly

## 5 Chase the sequence-parameter join coverage

The join between a scan and its parameters fires on 80.3% of segments. The other 19.7% fall back to the most recent earlier parameter row, which agrees with the truth only 28.5% of the time, and that fallback alone moves the tabular error from 9.7 s to 24.6 s. `03f_coverage_and_masking.py` is written, tested and has never been run — it answers this in about ten minutes with no Spark.

**Effort** ~10 min, notebook already exists **Fixes** F1 headroom

---

## **6 Reshape the duration tail and re-check the 13-hour day budget**

Truncation currently discards scheduled patients without surfacing it anywhere but a log line. Either the tail shrinks or the budget needs a louder failure mode; ideally both.

**Effort** small   **Fixes** F3

---

## **7 Give the handover model its long tail back**

Lowest priority. Visible on one dashboard chart, invisible everywhere else.

**Effort** small   **Fixes** F4

**Full cycle, in order**

Real data from `csv_pipeline/`, models and generation from `csv_pipeline_seqparams/`

- |    |                            |                                                        |
|----|----------------------------|--------------------------------------------------------|
| 01 | 01_exchange_preprocessing  | Real handover CSVs · zip download at the bottom        |
| 02 | 02_exam_preprocessing      | Real scan CSVs · read the coverage summary it prints   |
| 03 | 03_build_preprocessed_pkl  | Spark, ~4 min · builds the 414 MB training pickle      |
| 04 | 04_train_models            | GPU · run from the top so the pre-flight cells execute |
| 05 | 05_feature_analysis        | Permutation sweep · which inputs are alive             |
| 06 | 06_compare_models          | The acceptance gate · five numbered criteria           |
| 07 | 07_generate_synthetic_data | CPU · writes the synthetic CSVs per scanner            |
| 08 | qlik/consolidate.py        | Local · two files to drag into Qlik                    |

**Five traps, each of which has cost a full day**

- **Push before you run.** The notebooks clone `origin/main` from GitHub. Uncommitted local work simply does not exist to Databricks.
- **Never `%run` steps 01 or 02 from another notebook.** They begin with `%pip install`, which restarts the Python process and wipes the caller's state. It surfaces later as a misleading `NameError` in unrelated code.
- **Databricks caches modules in a live kernel.** Correct files on disk do not mean correct code running. Step 04 purges `sys.modules`; if you add a notebook, purge too, or call `dbutils.library.restartPython()`. Three consecutive retrains produced byte-identical losses before this was found.
- **Serving code takes its architecture from the checkpoint, never from `config.py`.** Read `MODEL_MANIFEST.json` and `protocol_vocab.json`. Omitting `num_protocols` is not an error — the checkpoint's protocol tensors are silently dropped as "unexpected" and every prediction falls back to the rare bucket. That ran unnoticed for three weeks.
- **A scanner with no exam CSV disappears three notebooks later.** That is why 21 configured serials produce 20 outputs — `155687` has no rows in the window, so step 03 writes no schedule for it and step 07 generates no days. Step 02 now prints which scanners fell out; read it.

**Refreshing the Qlik bundle**

Drop the per-scanner CSVs into `qlik/data/real/{exchange,exam}/` and `qlik/data/synthetic/{exchange,exam}/`, stripping the browser's `(1)` suffix —

plain `DATA_<serial>.csv` is the handover file, the `(1)` copy is the scan file. Run `python consolidate.py` , then drag the two files it writes into Qlik and hit reload. No chart edits, ever — the column names are prefixed deterministically so the six chart expressions never change.

The 20 synthetic scanners from the 31 August run are already sorted into place in the repository. The real side is waiting on the fresh 20-scanner extract.

08

Where everything lives

THE  
MAP

| PATH                                                    | WHAT IT IS                                                                                                                                     |
|---------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>AlternatingPipeline/</code>                       | The models, training loops, data preparation and validation metrics. Pure Python; runs and tests locally.                                      |
| <code>DatabricksPipeline/csv_pipeline/</code>           | Original pipeline. Owns the exchange and orchestration checkpoints and the real-data extracts.                                                 |
| <code>DatabricksPipeline/csv_pipeline_seqparams/</code> | The fork carrying the sequence-parameter work. This is where the examination model now lives.                                                  |
| <code>.../qlik/</code>                                  | Dashboard how-to, <code>consolidate.py</code> , and the landing zone for CSVs. The README carries the six chart specs and a graded scoreboard. |
| <code>tests/</code>                                     | 24 files, 395 tests, all green. Several encode a specific past outage so it cannot silently return.                                            |
| <code>presentation/</code>                              | The three-year results deck. <code>presentation/web/</code> is uncommitted.                                                                    |
| <code>docs/</code> , <code>_archive/</code>             | Architecture notes; superseded pipeline generations kept for reference.                                                                        |

Honest list of what is not done

No protocol-free accuracy number exists

Every error figure quoted anywhere, including the 19.1 s above, comes from a model that can see the protocol name. The number that decides whether this ships has never been measured.

03f has never been run

Written, unit-tested, needs no Spark, answers the join-coverage question in about ten minutes. It has simply never been executed on a cluster.

---

### The shipped CSVs lag the code

Two generator fixes behind: the intra-visit pause and the protocol sampler. Verified by measurement, not assumed.

---

### Uncommitted work in the tree

`qlik/README.md` and `qlik/consolidate.py` carry local edits (label normalisation across the two sides), and `presentation/web/` is untracked. Deliberate — none of it was mine to commit unreviewed.

---

**If I had one more month.** I would spend the first day on step 2 above — finding the real body-part field — because it is a one-line fix that unblocks the most visible failure in the dashboard, and it is the kind of thing that stays broken for a year because nobody believes it is that cheap. Then the rest of the month on F1, and only on F1. The parameter work is either worth 8 seconds of accuracy or it is worth nothing, and one honest ablation tells you which. Everything else on the list is a day's work that can wait.

Thanks for three good years. — Luke